SYNOPSIS

ON

“Distributed Storage Server Deployment on AWS”

Submitted By:

Team 1

Submitted To:

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1. Objective

The main objective of this project is to design and deploy a **Distributed Storage Server System** using cloud computing technologies. The system will be implemented using **Amazon Web Services (AWS)** to provide scalable, reliable, and highly available data storage.

Modern organizations generate large volumes of data that must be stored securely and accessed efficiently. Traditional storage systems often face limitations such as single points of failure, lack of scalability, and high maintenance costs.

This project aims to solve these problems by implementing a **distributed storage architecture** in which files are stored across multiple servers instead of a single system. By distributing the storage workload across several nodes, the system improves performance, ensures fault tolerance, and increases reliability.

The system will utilize cloud services such as **Amazon EC2**, **Amazon Elastic File System**, **Amazon RDS**, **AWS Elastic Load Balancing**, and **AWS CloudWatch** to build a robust distributed storage infrastructure.

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2. Features

The proposed distributed storage system provides the following important features:

1. Distributed Storage Architecture

Files are stored across multiple storage servers instead of a single server, which improves system reliability and performance.

2. High Availability

Multiple servers ensure that the system continues to function even if one server fails.

3. Scalability

The system can easily scale by adding additional storage nodes when data or user demand increases.

4. Shared File Storage

Using **Amazon Elastic File System**, multiple servers can access the same storage location simultaneously.

5. Load Balancing

AWS Elastic Load Balancing distributes user requests across servers to prevent overloading.

6. Metadata Management

File details such as file name, storage location, and upload time are stored in **Amazon RDS**.

7. Fault Tolerance

If a storage node fails, requests are automatically redirected to other active nodes.

8. Monitoring and Logging

System performance is monitored using **AWS CloudWatch**.

9. Secure Access

Security groups and access policies ensure that only authorized users can access stored data.

10. Cost Efficiency

Cloud resources allow organizations to pay only for the infrastructure they use.

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3. Scope

The scope of this project focuses on designing and implementing a distributed storage infrastructure using cloud computing technologies. The system will be developed using services provided by Amazon Web Services, which allows organizations to store and manage large volumes of data efficiently in a cloud environment.

The project mainly involves deploying multiple Amazon EC2 instances that will function as storage servers. These servers will work together to handle user requests for uploading and retrieving files. To ensure that all servers can access the same files, a shared storage system will be implemented using Amazon Elastic File System. This shared storage will allow the distributed servers to read and write data from a centralized location.

To manage incoming traffic efficiently, the system will use AWS Elastic Load Balancing, which distributes user requests across multiple servers and prevents overloading of a single node. In addition, a relational database service such as Amazon RDS will be used to store metadata information related to stored files, including file names, storage paths, and timestamps. System monitoring will also be implemented using AWS CloudWatch to track server performance and resource utilization.

However, the project focuses mainly on demonstrating the basic concept of distributed storage in a cloud environment. Advanced enterprise-level security mechanisms and large-scale production deployment are outside the scope of this project.

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4. Methodology

The methodology of this project involves designing and implementing a distributed cloud storage architecture by integrating several cloud services. The development process begins with planning the system architecture and identifying the required cloud resources. An account will be created on Amazon Web Services, and the necessary infrastructure components will be configured.

The first step in the implementation process involves deploying multiple Amazon EC2 instances that will act as storage nodes. These instances will handle user requests and perform file storage operations. To enable shared file access among the servers, Amazon Elastic File System will be configured and mounted on each EC2 instance. This shared storage system allows multiple servers to access the same data simultaneously.

After setting up the storage environment, a load balancing mechanism will be implemented using AWS Elastic Load Balancing. The load balancer will distribute incoming client requests among the available servers to ensure balanced workload and improved system performance. To maintain information about stored files, a database will be created using Amazon RDS, where metadata such as file names, locations, and timestamps will be stored.

Finally, system monitoring will be configured using AWS CloudWatch, which will track the health and performance of the deployed servers. Testing and validation will then be performed to ensure that the distributed storage system functions correctly and efficiently.

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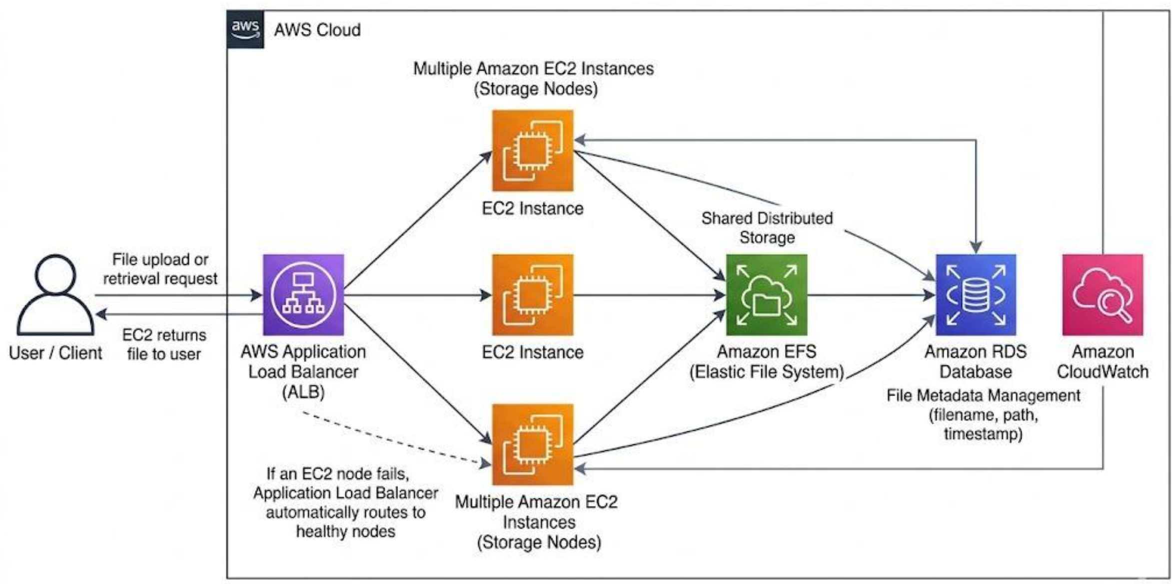
5. System Architecture

The system architecture consists of several components that work together to provide distributed storage.



Working

1. Users upload files through the application interface.
2. The request is received by the **AWS Elastic Load Balancing** service.
3. The load balancer forwards requests to available **Amazon EC2** instances.
4. Files are stored in **Amazon Elastic File System**.
5. Metadata information is stored in **Amazon RDS**.
6. System performance is monitored using **AWS CloudWatch**.

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6. Implementation Plan

Week No.	Task	Mentor Evaluation
Week 1 – 2	Requirement analysis, project planning, and initial setup of the cloud environment using **Amazon Web Services**. IAM users and permissions will be configured.	Done / Not Done
Week 3 – 4	Setup of network infrastructure including VPC configuration and deployment of **Amazon EC2** instances that will act as storage servers.	Done / Not Done
Week 5 – 6	Implementation of shared storage using **Amazon Elastic File System** and integration with EC2 instances.	Done / Not Done
Week 7 – 8	Configuration of **AWS Elastic Load Balancing** and setup of metadata database using **Amazon RDS**.	Done / Not Done
Week 9 – 10	Monitoring setup using **AWS CloudWatch**, system testing, debugging, and final documentation.	Done / Not Done

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7. Expected Outcomes

- Successful deployment of a distributed storage system on the cloud.
- Improved reliability and fault tolerance in data storage.
- Efficient file storage and retrieval across multiple servers.
- Balanced workload using load balancing techniques.
- Continuous monitoring of system performance.

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8. Future Scope

The system can be improved in several ways in the future.

- Integration of advanced security mechanisms such as encryption and multi-factor authentication.
- Automatic scaling of servers using cloud auto-scaling features.
- Implementation of backup and disaster recovery systems.
- Integration with container technologies such as **Docker** and orchestration tools like **Kubernetes**.
- Performance optimization using caching and distributed data processing.

These enhancements would make the system more efficient and suitable for large-scale enterprise applications.

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9. Work Distribution Among Team Members

Sumit chaudhary

responsible for overall project planning and initial system setup. This includes creating and configuring the cloud environment on **Amazon Web Services**, setting up user permissions, also coordinate the activities of the team and ensure that the project follows the planned schedule.

Nikhil Sharma

handle the deployment and configuration of virtual servers using **Amazon EC2**,includes launching EC2 instances, configuring operating systems, and setting up necessary software for storage services.

Samman Parashar

will implement the distributed storage environment using **Amazon Elastic File System**. This include configuring shared storage and ensuring that all EC2 instances can access and store files within the shared file system.

Raj Yadav

will manage the database and traffic distribution system, includes creating and configuring the metadata database using **Amazon RDS** and setting up **AWS Elastic Load Balancing** to distribute incoming requests among servers.

Vishal

will be responsible for monitoring system performance using **AWS CloudWatch**, performing system testing, debugging any issues, and preparing the final project documentation and report.

10. Reference Books

1. Rajkumar Buyya – *Cloud Computing: Principles and Paradigms*.
2. Thomas Erl – *Cloud Computing: Concepts, Technology & Architecture*.
3. Michael Miller – *Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online*.
4. Official documentation of **Amazon Web Services**.

11. Conclusion

The **Distributed Storage Server Deployment on AWS** project demonstrates how cloud computing technologies can be used to design a reliable and scalable data storage system. By using services provided by **Amazon Web Services**, the project shows how data can be stored across multiple servers to improve system performance, reliability, and availability.

The implementation of services such as **Amazon EC2** for compute resources, **Amazon Elastic File System** for shared storage, **Amazon RDS** for metadata management, and **AWS Elastic Load Balancing** for traffic distribution helps in building an efficient distributed architecture. Additionally, system monitoring through **AWS CloudWatch** ensures better management and performance tracking of the infrastructure.

Overall, this project highlights the advantages of distributed storage systems in handling large volumes of data while maintaining high availability and fault tolerance. It also provides practical knowledge of deploying and managing cloud infrastructure, which is highly relevant in modern computing environments.